

DA5400: Foundations of Machine Learning

Topic 3: Regression – Part 1



Outline

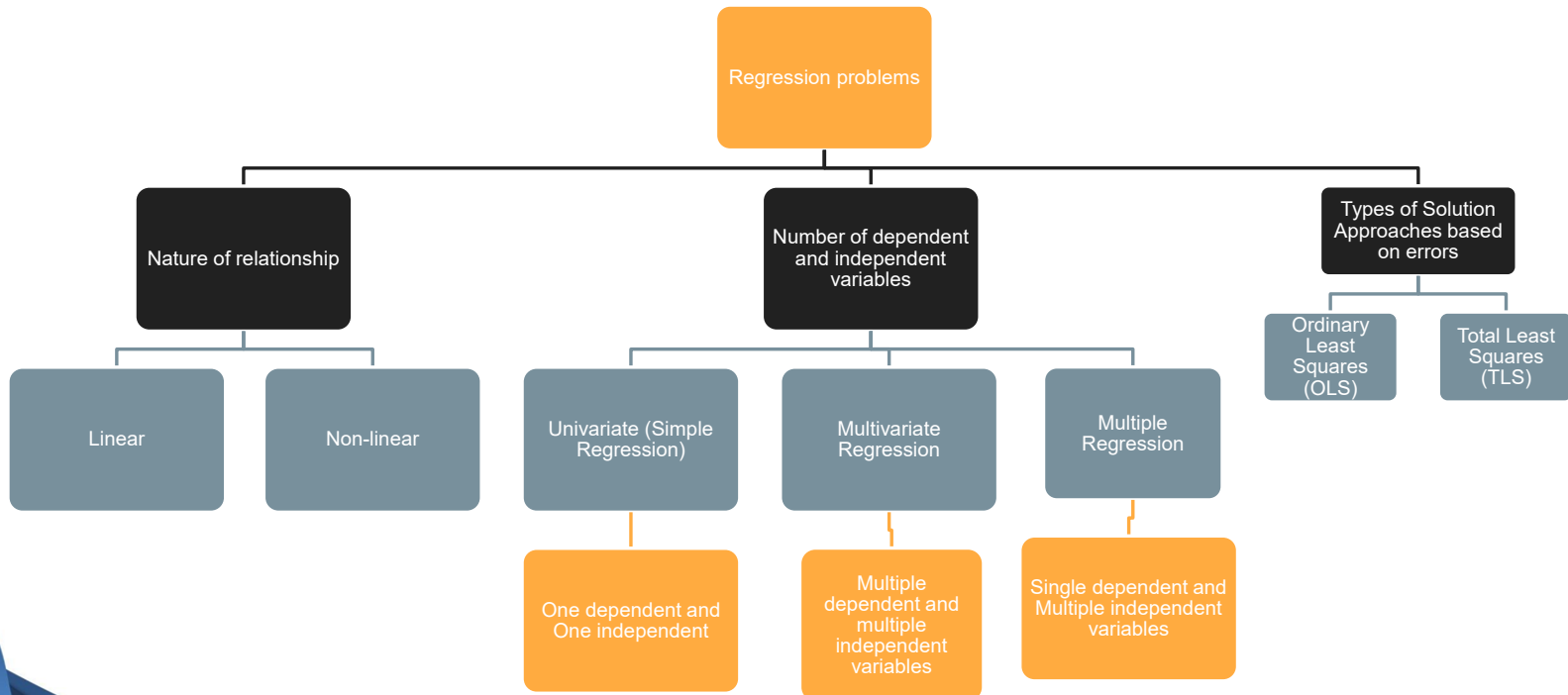
- Introduction to Regression
- Types of Regression problems
- Linear Regression
- Simple and Multiple Linear Regression
- OLS Solution for Linear Regression
- TLS and WOLS/WTLS Solutions for Linear Regression
- Other forms of Linear Regression
- Multi-Variate Linear Regression

Regression

Introduction to Regression

- Refers to an approach in which relationships (model) are identified between a set of dependent (predicted or regressed or output) variables and a set of independent (predictor or regressor or input) variables
- Dependent variables should be continuous numerical variables
- Applications of regression in predictive tasks:
 - Regression model to predict price of a house given size, location, # bedrooms, etc.
 - Regression model to predict blood pressure given age, weight, activity level, diet, etc.
 - Regression model to predict sales of a product given advertising budget, price, customer demographics, etc.
 - Regression model to predict strain on a material given the force applied, material properties, etc.

Types of Regression Problems



Linear Regression

True Vs Observed Vs Predicted Value

- Value of Dependent variable in regression can take 3 different values in a sample
- **True value (y^t):**
 - Refers to the ground truth or actual value that is perfectly related to the independent variable
 - Generally unknown in most practical examples
- **Observed value (y):**
 - Refers to the measured or observed value available in the dataset
 - Generally, deviates from the true value due to random noise
- **Predicted value ($\hat{y}$):**
 - Refers to the value predicted or estimated by a regression model
 - Can deviate from the observed value

Linear Regression

- Relation between the dependent and independent variables is assumed to be a linear function of the parameters (regression coefficients)
- **Note:** Function need not be linear in the variables
- Different types of linear regression problems:
 1. **Simple Linear Regression:** Single dependent variable and single independent variable related linearly

$$y(x, \mathbf{w}) = w_0 + w_1 x + \epsilon$$

2. **Multiple Linear Regression:** Single dependent variable and multiple independent variables related linearly

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_D x_D + \epsilon$$

Generalisation – Linear combination of functions of variables

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1 \phi_1(\mathbf{x}) + w_2 \phi_2(\mathbf{x}) + \dots + w_D \phi_D(\mathbf{x}) + \epsilon$$

Linear Regression

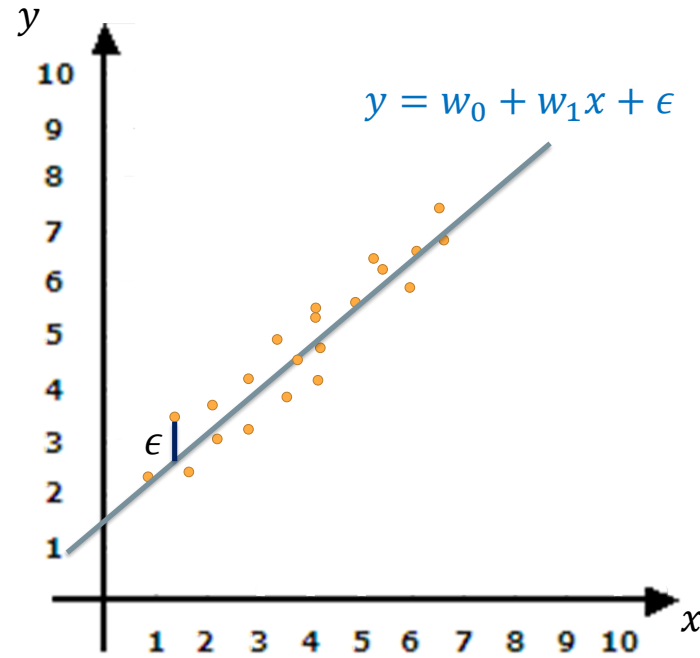
3. **Multivariate Linear Regression:** Multiple dependent variables and multiple independent variables related linearly

$$\begin{aligned}y_1(\mathbf{x}, \mathbf{w}) &= w_{01} + w_{11}x_1 + w_{21}x_2 + \cdots + w_{D1}x_D + \epsilon_1 \\y_2(\mathbf{x}, \mathbf{w}) &= w_{02} + w_{12}x_1 + w_{22}x_2 + \cdots + w_{D2}x_D + \epsilon_2 \\&\vdots \\y_m(\mathbf{x}, \mathbf{w}) &= w_{0m} + w_{1m}x_1 + w_{2m}x_2 + \cdots + w_{Dm}x_D + \epsilon_m\end{aligned}$$

Simple and Multiple Linear Regression

- Considering n independent samples of multi-variate data, the relations between the samples can be written in matrix form as:

$$\mathbf{y} = [\mathbf{1}_n \quad \mathbf{X}] \mathbf{w} + \boldsymbol{\epsilon}$$
$$\mathbf{y} = \mathbf{X}^* \mathbf{w} + \boldsymbol{\epsilon}$$



Random Error ϵ

- Error variable ϵ accounts for the random variability in the dependent variable that cannot be explained by the independent variables
- Difference between the actual value and predicted value of the dependent variable
- Error accounts for the following factors:

Random noise: Measurements using instruments or sensors are not perfect and there can be human errors in noting measurements

- Eg: Voltage Vs Current: $V = IR$ is not perfectly observed in measurements

Assumptions on Random Error ϵ

- In general, the following assumptions are made on the random error:
 - $\mathbb{E}[\epsilon] = 0$
 - Expectation of error term should be 0
 - In other words, actual values of the output should be equally spread around than predicted value
 - Predicted values should NOT be consistently over or under-estimating the actual values of output
 - Homoscedastic errors: $Cov[\epsilon] = \sigma^2 I$
 - Variance of error term should be same for all values of independent variables and errors across different observations should not be correlated
 - Eg: Error variance in house price prediction should be same for economical as well as expensive houses
 - Errors should be normally distributed

Assumptions on Random Error ϵ

- In general, the following assumptions are made on the random error:
 - Independent variables and error terms should not be correlated
 - Eg: Output: Wages, Input: Years of Education
 - Here, talent and ability might be an important factor in predicting wages which is not considered
 - People higher talent will get higher wages with same years of education
 - Positive error
 - People with higher talent tend to spend more years in education because they find it easier to study
 - So, there is a chance for correlation between error and years of education

Ordinary Least Squares Solution

Finding a Linear Regression Model from Data

- **Model Structure:** $\hat{y} = w_0 + w_1x_1 + \dots + w_Dx_D$
- **Known:** Mapped Input ($\mathbf{X}$) and Output Data ($\mathbf{y}$)
- **Objective:** To estimate regression coefficients $\mathbf{w}$ which minimize the errors in prediction
- **Assumptions:**
 - **No Multi-collinearity:** Independent variables should not be highly correlated
 - Number of samples in the dataset should be greater than the number of regression coefficients
 - **OLS Assumption:** $\mathbf{X}$ is observed without noise
- **Note:** Estimated regression coefficients will be different from true regression coefficients

Least Squared Approach (OLS)

- **Residual:** Difference between observed output and predicted output of a regression model

- Sum of squared residuals in predicting y is given by:

$$SSR = (\mathbf{y} - \mathbf{X}^* \mathbf{w})^T (\mathbf{y} - \mathbf{X}^* \mathbf{w})$$

- Least squares estimate of $\mathbf{w}$ are the coefficients that minimize SSR

$$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} SSR$$

- **Result:** Analytical solution to the least squares estimate of $\mathbf{w}$ is given by:

$$\hat{\mathbf{w}} = (\mathbf{X}^{*T} \mathbf{X}^*)^{-1} \mathbf{X}^{*T} \mathbf{y} - \text{Normal Equation}$$

OLS Solution = MLE

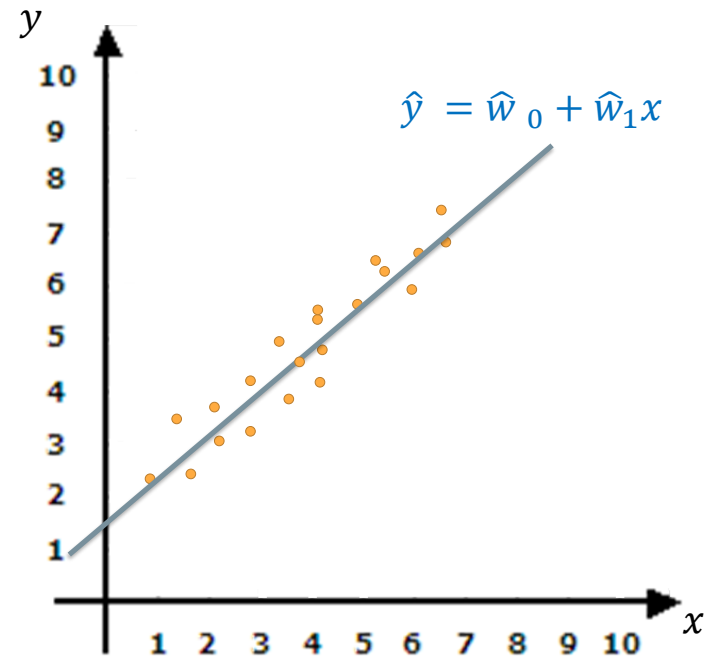
Interpretation of $\hat{\mathbf{w}}$

- Regression coefficients $\hat{\mathbf{w}}$ best fit the available data
- Best fit implies $\hat{\mathbf{w}}$ is unbiased estimate of $\mathbf{w}$
- **Note:** It is the best fit under the assumptions made. If any of the assumptions fail, OLS solution would be biased
- **Interpretation:** Each coefficient w_i is the change in the dependent variable for unit change in the independent variable x_i , holding everything else constant

Linear Algebraic Interpretation of OLS

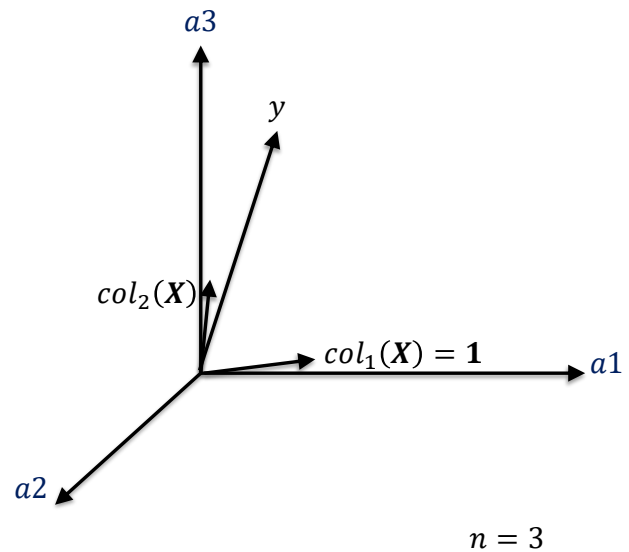
Vector Spaces Associated with a Data Matrix

- **Variables as dimensions:**
 - **Dimensions:** Each dimension or axis of the vector space represents a variable
 - **Vectors:** Each vector in the space (row in data matrix) represents a sample across all variables



Vector Spaces Associated with a Data Matrix

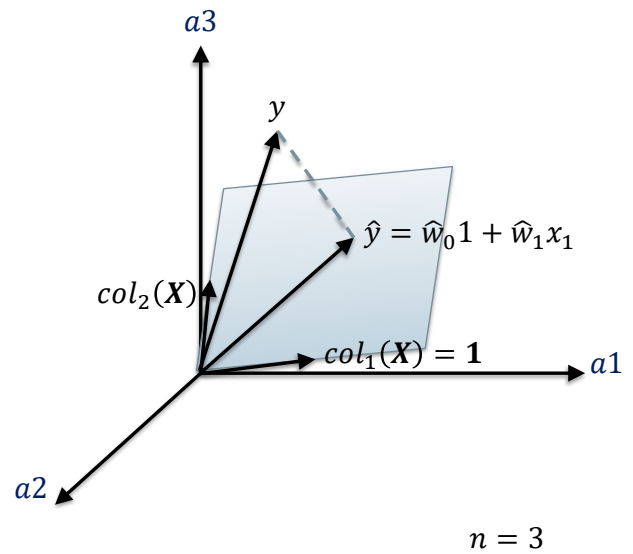
- **Variables as dimensions:**
 - **Dimensions:** Each dimension or axis of the vector space represents a variable
 - **Vectors:** Each vector in the space (row in data matrix) represents a sample across all variables
- **Samples as dimensions:**
 - **Dimensions:** Each dimension or axis of the vector space represents a sample
 - **Vectors:** Each vector in the space (column in data matrix) represents a variable with all its sample values



Linear Algebraic Interpretation of OLS

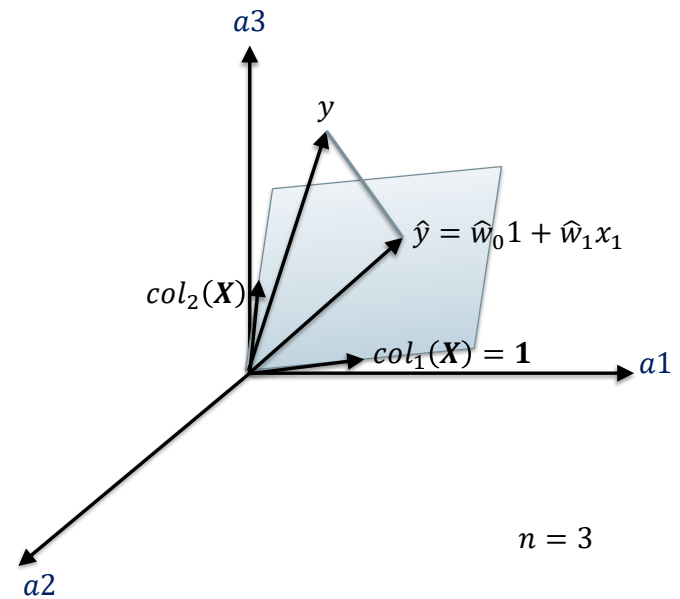
- OLS Solution in matrix form: $\hat{\mathbf{y}} = \mathbf{X}^* \hat{\mathbf{w}}$
- Interpretation:
 - $\hat{\mathbf{y}}$ lies in the column space of $\mathbf{X}^*$
 - $\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the column space of $\mathbf{X}^*$
- Projection Matrix:

$$\begin{aligned}\hat{\mathbf{y}} &= \mathbf{X}^* \hat{\mathbf{w}} \\ \hat{\mathbf{y}} &= \mathbf{X}^* (\mathbf{X}^{*T} \mathbf{X}^*)^{-1} \mathbf{X}^{*T} \mathbf{y} \\ \hat{\mathbf{y}} &= \mathbf{P} \mathbf{y}\end{aligned}$$



Linear Algebraic Interpretation of OLS

- Residual vector (Actual – Predicted):
$$\mathbf{r} = \mathbf{y} - \mathbf{X}^* \hat{\mathbf{w}}$$
- Result:** Residual vector is perpendicular to the column space of $\mathbf{X}^*$
- Proof:**
$$\begin{aligned}\mathbf{X}^{*T} \mathbf{r} &= \mathbf{X}^{*T} (\mathbf{y} - \mathbf{X}^* \hat{\mathbf{w}}) \\ &= \mathbf{X}^{*T} \mathbf{y} - \mathbf{X}^{*T} \mathbf{X}^* (\mathbf{X}^{*T} \mathbf{X}^*)^{-1} \mathbf{X}^{*T} \mathbf{y} = \mathbf{0}\end{aligned}$$



Summary

- Residuals capture the error in prediction in an estimated regression model
- Regression model is obtained by estimating the coefficients which minimize the residuals
- Analytical OLS solution to the regression problem is given by the normal equation

$$\hat{\mathbf{w}} = (\mathbf{X}^{*T} \mathbf{X}^*)^{-1} \mathbf{X}^{*T} \mathbf{y}$$

- Regression coefficients capture the change in the dependent variable for unit change in the independent variables
- OLS prediction of the dependent variable is the projection of vector of actual values onto the column space of the data matrix

Total Least Squares Solution (TLS)

Linear Algebraic Concepts for TLS

Singular Value Decomposition (SVD)

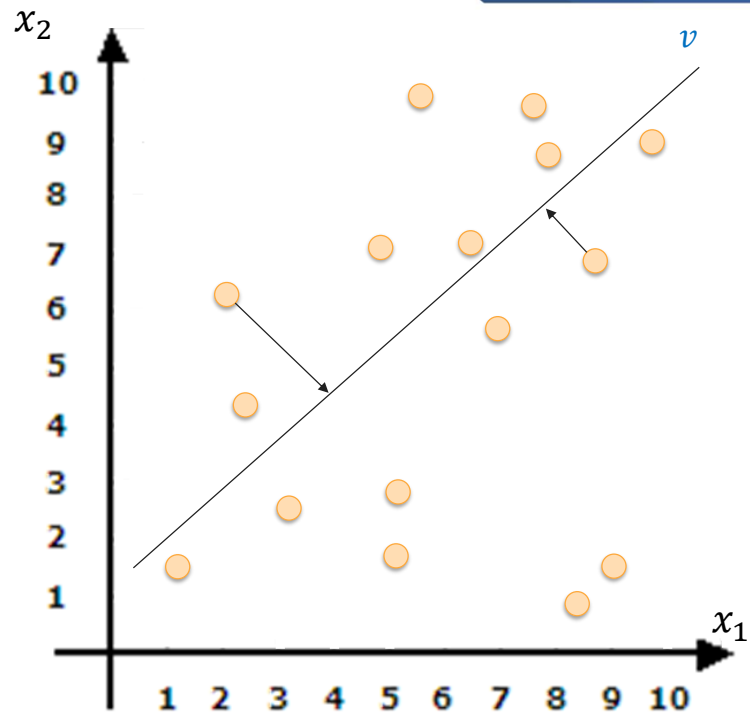
- Any $m \times n$ matrix X can be factorized (decomposed) into three matrices based on singular values and singular vectors

$$X_{m \times n} = U_{m \times m} S_{m \times n} V_{n \times n}^T$$

- Columns of U : Left singular vectors u
- Columns of V : Right singular vectors v
- Matrix S has singular values (σ) of X along the main diagonal

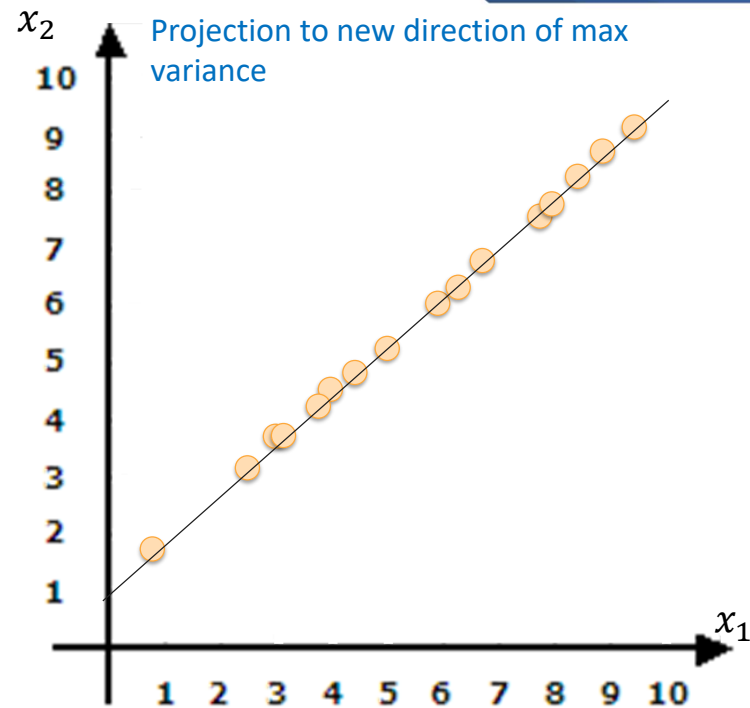
Interpretation of Singular Values & Singular Vectors of a Data Matrix

- Suppose X is a data matrix with m rows (samples) and n columns (features)
- Interpretation of Singular values and singular vectors:
 - Right singular vectors v : directions along which the variance in the data is σ
 - Left singular vectors u : projections of data samples onto the directions given by v
- **Result:** Right singular vector + largest singular value = direction of maximum variance in the data



Interpretation of Singular Values & Singular Vectors of a Data Matrix

- Suppose X is a data matrix with N rows (samples) and D columns (features)
- Interpretation of Singular values and singular vectors:
 - Right singular vectors v : directions along which the variance in the data is σ
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Sum of r Rank 1 Matrices

- **Result:** A matrix of rank r can be written as a sum of r matrices each having rank 1
- This can be written by using the r non-zero singular values and their corresponding singular vectors
- Consider a matrix X of rank r such that $r < n < m$

$$X = U S V^T$$

$$= \begin{bmatrix} u_1 & u_2 & \dots & u_m \end{bmatrix}_{m \times m} \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_r & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & 0 \end{bmatrix}_{m \times n} \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{bmatrix}_{n \times n}$$

$$= u_1 \sigma_1 v_1^T + u_2 \sigma_2 v_2^T + \dots + u_r \sigma_r v_r^T$$

$$[m \times 1] [1 \times 1] [1 \times n] \rightarrow m \times n$$

Low Rank Approximation

- A matrix $\mathbf{X}$ of rank r can be approximated by a matrix $\mathbf{X}_l$ of rank $l < r$ by taking the largest l singular values and the corresponding singular vectors

$$\mathbf{X}_l = \mathbf{u}_1\sigma_1\mathbf{v}_1 + \mathbf{u}_2\sigma_2\mathbf{v}_2 + \cdots + \mathbf{u}_l\sigma_l\mathbf{v}_l$$

➤ Example:

$$\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0.7 & -0.7 \\ 1 & 0 & 0 \\ 0 & 0.7 & 0.7 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1.4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Frobenius Norm of a Matrix

- Measure which gives a single number representing the size or magnitude of a matrix
- Mathematically, it is the square root of the sum of the absolute squares of its elements

$$||\mathbf{X}||_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n X_{ij}^2}$$

- In other words, Frobenius norm vectorizes the matrix and then computes its magnitude

TLS Solution to Linear Regression

TLS Approach

- **Recall:** In the OLS approach, errors are assumed to be present only in the dependent variable but not the independent variable
- **TLS approach:** Errors in both independent and dependent variables are minimized while obtaining a model

- True model:

$$y^t = w_0 + w_1 x^t$$

- Observed model:

$$y = w_0 + w_1 x + (w_1 \epsilon_x - \epsilon_y)$$

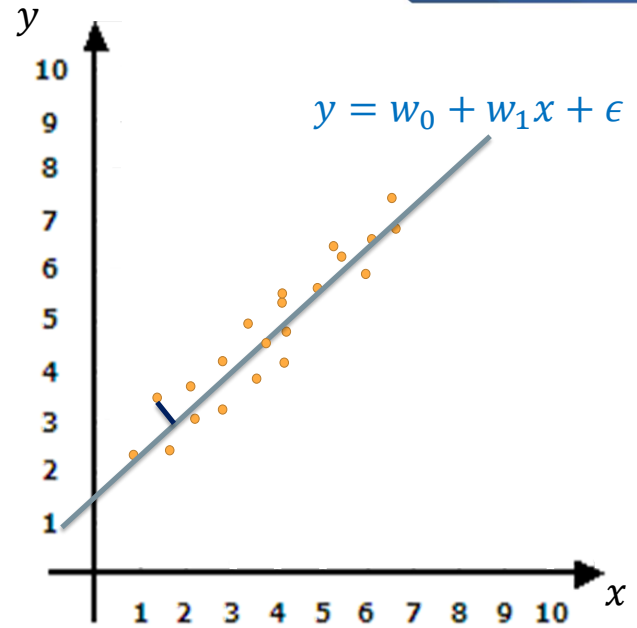
- Values of both dependent and independent values are to be estimated in this case

Motivation for TLS Approach

- Mathematical objective becomes:

$$\begin{aligned} \min_w & \| [X, y] - [\hat{X}, \hat{y}] \|_F \\ \text{s.t. } & \hat{y} \in \text{Eolspace}(\hat{X}) \\ & \hat{y} = \hat{X} \hat{w} \end{aligned}$$

- From a vector space perspective, the sum of perpendicular distances is minimized.
- No closed form solution exists. A numerical solution based on SVD is developed



TLS Solution Steps

- **Step 1:** Scale the data pertaining to all independent and dependent variables to zero mean and unit variance
 - Singular values of a data matrix depend on the scale of the variables
 - Since singular values are used in the solution, this scaling ensures all variables are equally treated by
- **Step 2:** Construct the augmented data matrix combining the input and output data matrices

$$X_{aug} = [X \quad y]_{N \times (D+1)}$$

- Note: No intercept is assumed by default but can be included if required
- **Step 3:** Perform SVD on the augmented data matrix

$$X_{aug} = USV^T$$

TLS Solution Steps

- **Step 4:** Identify the smallest singular value and its corresponding right singular vector
 - σ_{D+1} is the smallest singular value with right singular vector v_{D+1}
- **Step 5:** Partition the singular vector v_{D+1} corresponding to independent and dependent variables as follows:

$$v_{D+1} \in \mathbb{R}^{D+1}$$
$$v_{D+1} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$
$$v_x \rightarrow n \times 1$$
$$v_y \rightarrow 1 \times 1$$

TLS Solution Steps

- **Step 6:** Estimates for regression coefficients are computed as follows:

$$\hat{\mathbf{w}} = \frac{-\mathbf{1}}{\mathbf{v}_y} [\mathbf{v}_x]$$

- **Intuition:** $\mathbf{v}_{D+1}$ is the direction perpendicular to the direction of maximum variance in the data
- **Step 7:** Get corrected estimates for the data matrices $\mathbf{X}$ and $\mathbf{y}$ by taking the low rank matrix approximation of the augmented data matrix

$$\mathbf{X}_{aug}^L = \mathbf{u}_1 \sigma_1 \mathbf{v}_1^T + \mathbf{u}_2 \sigma_2 \mathbf{v}_2^T + \dots + \mathbf{u}_D \sigma_D \mathbf{v}_D^T$$

- **Note:** Interpretation of TLS coefficients remains same as OLS coefficients

Summary

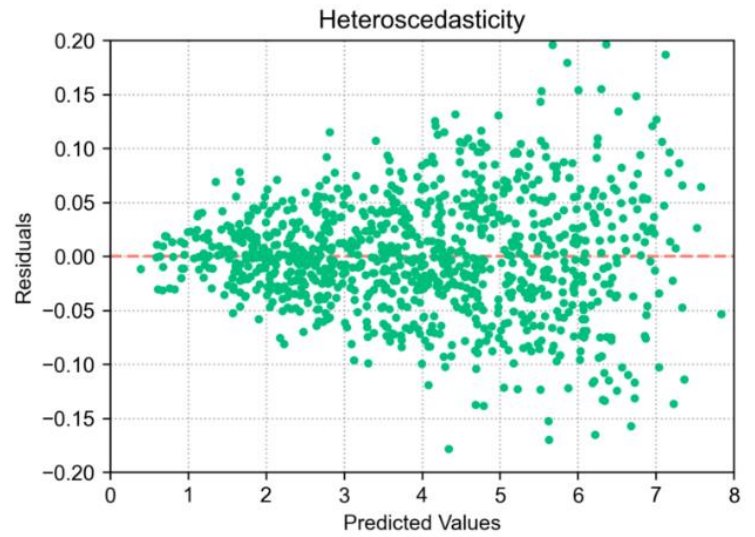
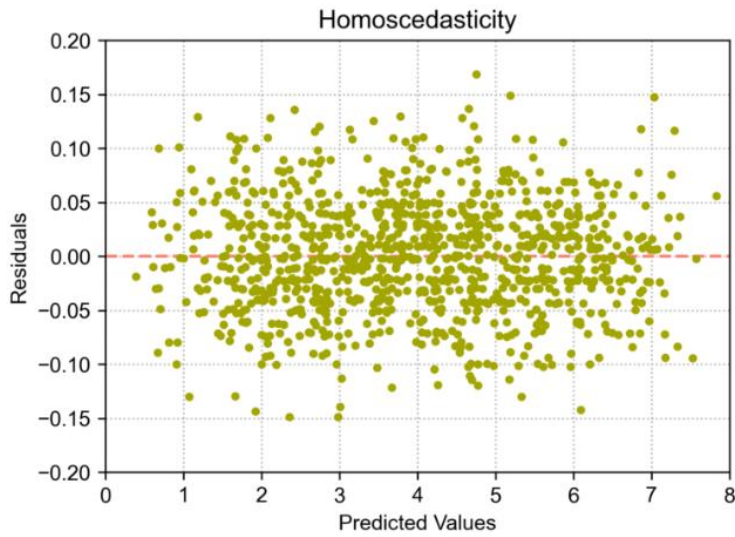
- Right Singular vectors of a data matrix can be interpreted as directions along which variance equal to singular values is captured
- Every matrix can be approximated by a lower rank r matrix by taking the largest r singular values and corresponding singular vectors
- Low rank matrix approximation has different applications including TLS solution
- TLS regression is an approach where errors in both independent and dependent are minimized while obtaining the regression coefficients
- TLS solution is obtained by taking SVD of the augmented data matrix

Weighted OLS and TLS

Reliability of Measurements

- In practice, not all measurements are equally reliable
- Differences in reliability arise due to:
 - Data collection process
 - Measurement instruments
 - Experimental or operational conditions
- In other words, error variance differs across observations:
$$\text{var}(\epsilon_i) = \sigma_i^2$$
- This leads to heteroscedasticity
- **Impact:** Standard OLS/TLS are no more estimators with least variance among unbiased estimators
- **Solution:** Weighted OLS/TLS resolves this by factoring the heteroscedasticity in noise while estimating the parameters

Reliability of Measurements



Source: towardsdatascience.com

Weighted OLS

- Can be used if the value of dependent variable in some samples is known to be more reliable or less error prone compared to others
- Objective of Weighted OLS problem is modified as follows:

$$\min_w \sum_{i=1}^N \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}$$

- Analytical solution to Weighted OLS problem:

$$\begin{aligned}\hat{\mathbf{w}} &= (\mathbf{X}^{*T} \mathbf{\Sigma}^{-1} \mathbf{X}^*)^{-1} \mathbf{X}^{*T} \mathbf{\Sigma}^{-1} \mathbf{y} \\ \mathbf{\Sigma} &= \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_N^2)\end{aligned}$$

Interpretation of Weighted OLS

- Weight matrix A adjusts the influence of each sample on the regression model
- In other words, the samples which are more reliable or have less error variance will have more impact in estimating the coefficients
- Different approaches exist depending on the availability of information on the samples

Weights for Weighted OLS

- **Approach 1: Reliability based Weights**
 - When data comes from multiple sources and one source is known be more reliable than the other
 - Sources with higher reliability can be given higher weight
- **Approach 2: Iterative Re-weighting**
 - Initial weights are estimated by performing standard OLS
 - Then WOLS is performed and weights are re-estimated
 - This goes iteratively until the weights converge

$$OLS \rightarrow \alpha_i = \frac{1}{\sigma_i^2} \approx \frac{1}{(y_i - \hat{y}_i)^2} \rightarrow WOLS$$


Weighted TLS

- Used when different samples have different overall reliability
- Errors are present in both input and output variables
- Within a sample, all variables are assumed to have the same noise variance
- Steps 2 and 3 of the TLS approach get modified to get Weighted TLS
- **Step 2:** Construct the augmented data matrix combining the input and output data matrices

$$X_{aug}^s = \Sigma^{\frac{1}{2}} [X \quad \mathbf{y}]_{N \times (D+1)}$$

- **Step 3:** Perform SVD on the augmented data matrix

$$X_{aug}^s = USV^T$$

- Remaining steps are performed as Standard TLS

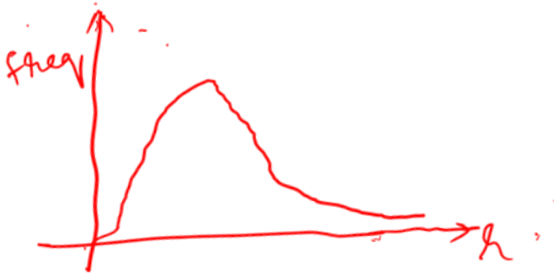
Other Forms of Linear Regression

Other forms of Linear Regression

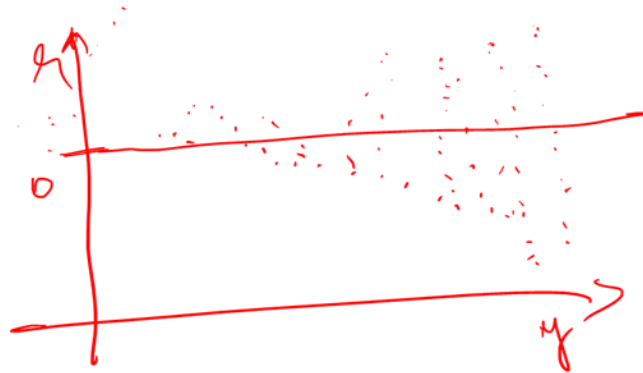
- Q: What if relation between observed dependent and independent variables is not linear?
- Recall: Linear regression assumes linearity in terms of coefficients but not variables
- Q: Is the relation between the dependent variable or its transformed form and independent variables or their transformed form linear?
- In such cases, linear regression can be used by transforming the dependent or independent variables suitably
- Such transformations include:
 - Polynomial transformation – Polynomial Regression
 - Log transformation – Logarithmic Regression

Other forms of Linear Regression

- Q: When to try other forms of linear regression?
- Following are some of the scenarios:
 - Non-linear structure of the model is known
 - Residuals appear skewed violating the normality assumption – Histogram of residuals can be checked
 - Some non-random pattern is observed in the residuals vs output plot violating equal variance assumption



Regression



Polynomial Regression

- Independent variables in the linear regression model are n^{th} degree polynomials of the observed variables

$$y = w_0 + w_1x_1 + w_2x_1^2 + w_3x_2 + w_4x_2^2 + w_5x_1x_2$$

- OLS approach can be used to find the regression coefficients
- **Note:** Overfitting can be an issue because many terms get added with every additional degree of polynomial
- Degree of the polynomial can be decided using cross-validation with regression metrics

Logarithmic Regression

- Log transformation can be both in dependent and independent variables depending on the requirement
- Visualising log transformation:

x Age	y Income	$\ln(y)$
25	20,000	9.9
30	24,000	10
35	30,000	10.3
40	40,000	10.6
45	52,000	10.9
50	65,000	11.08
55	80,000	11.3



Logarithmic Regression

- Log transformation can be both in dependent and independent variables depending on the requirement
- Scenarios when Log transformation can be applied:
 1. **Skewed distribution:**
 - Variable has a skewed distribution
 - Skewness could be due to noise being proportional to signal: ($y = x\epsilon$)
 - Skewness could be due to heteroscedasticity
 - Log transformation makes noise additive and homoscedastic
 - Eg: Income or house prices which are generally right skewed

2. **Known Exponential Relation:**

$$y = ae^{bx}$$
$$\ln(y) = \ln(a) + bx$$

$$y_1 = w_0 + w_1x$$

- Eg: Population growth or radioactive decay models

Logarithmic Regression

- Scenarios when Log transformation can be applied:
 3. **Variance in error is not constant:**
 - Log transformation on dependent variable can stabilise variance in some cases
 - Eg: Variability in income may increase with level of income
 4. **Wide range:**
 - Variables which have a wide range
 - Applying log transformation compresses the scale and reduces the impact of large values on loss function

Logarithmic Regression

- Scenarios when Log transformation can be applied:

5. Percentage Changes:

- In many applications, the interest is in relative (percentage) changes, not absolute changes
- Eg: Modelling the % change in sales (dependent) as a function of advertising spends (independent variable)
- Log transformation on that variable can be applied

$$\frac{d}{dy}(\ln y) = \frac{1}{y}$$

$$d(\ln y) = \frac{dy}{y} \Rightarrow \ln y_2 - \ln y_1 \approx \frac{y_2 - y_1}{y_1}$$

- A small change in $\ln y$ corresponds to a relative (or percentage) change in y
- Eg: If $\ln y$ increases by 0.01, then y increases by 1%

Logarithmic Regression

- Consider a **log-linear model**:

$$\ln y = w_0 + w_1 x$$

- Differentiate w.r.t x :

$$\frac{1}{y} \frac{dy}{dx} = w_1 \Rightarrow \frac{dy}{y} = w_1 dx$$

- **Implies**: A unit increase in x implies approx. $w_1 \times 100\%$ change in y

- Consider a **linear-log model**:

$$y = w_0 + w_1 \ln x$$

- Differentiate w.r.t x :

$$\frac{dy}{dx} = \frac{w_1}{x} \Rightarrow dy = w_1 \frac{dx}{x}$$

- **Implies**: A 1% increase in x implies approx. change of $w_1/100$ units in y

Logarithmic Regression

- OLS approach can be used to find the regression coefficients
- **Points to Note:**
 - OLS related assumptions should be reasonably satisfied after performing log transformation
 - Log transformation can be applied only for variables taking positive values
 - Shifting can be done in case of 0 values
- **Caution:** If the range of x is small, there could be high correlation between x and $\ln x$ | x and x^2

Multi-Variate Linear Regression

Multi-Variate Linear Regression

- Multiple dependent variables and multiple independent variables related linearly

$$y_1 = \beta_{01} + \beta_{11}x_1 + \beta_{21}x_2 + \dots + \beta_{p1}x_p + \epsilon_1$$

$$y_2 = \beta_{02} + \beta_{12}x_1 + \beta_{22}x_2 + \dots + \beta_{p2}x_p + \epsilon_2$$

$\vdots$

$$\mathbf{Y}_{N \times m} = \mathbf{X}_{N \times (D+1)} \mathbf{W}_{(D+1) \times m} + \mathbf{E}_{N \times m}$$

$$y_m = \beta_{0m} + \beta_{1m}x_1 + \beta_{2m}x_2 + \dots + \beta_{pm}x_p + \epsilon_m$$

- Assumptions:
 - Mostly same assumptions as multiple linear regression
 - Errors are independent across samples of an output but error covariance across outputs is allowed
 - Errors for each output, across samples are jointly normal (multi-variate normal)

Two Solution Approach

- OLS solution Multi-Variate Linear regression can be applied in two ways:
 - Multiple Inputs Single Output (MISO)
 - Multiple Inputs Multiple Outputs (MIMO)
- **MISO Approach:**
 - In this approach, the model is broken down into multiple models with each having only 1 dependent variable
 - Each model can be obtained just like any multiple linear regression model
$$\widehat{\beta}_i = (X^{*T} X^*)^{-1} X^{*T} y_i$$
 - This approach can be applied only when the following assumptions are satisfied:
 - Dependent variables are independent of one another
 - Error terms pertaining to different dependent variables should not be correlated

Two Solution Approach

- OLS solution Multi-Variate Linear regression can be applied in two ways:
 - Multiple Inputs Single Output (MISO)
 - Multiple Inputs Multiple Outputs (MIMO)
 - **MIMO Approach:**
 - In this approach, a single model is built which models all relationships
 - Normal equation works in this case as well
- $$\hat{\mathbf{B}} = (\mathbf{X}^{*T} \mathbf{X}^*)^{-1} \mathbf{X}^{*T} \mathbf{Y}$$
- This model accounts for any dependence that exists between the dependent variables and also the error terms